**EE4-01**
**EE9-SC2**

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2015

MSc and EEE/EIE PART IV: MEng and ACGI

# ADVANCED COMMUNICATION THEORY

Corrected Copy

Wednesday, 29 April 2015, 10:00 am

Time allowed: 3:00 hours

**There are 26 questions on this paper.**

**Answer ALL questions.**

**The multiple choice questions together account for 40% of the marks.**

*Answers to multiple choice questions 1-20 should be given on the paper itself.*

*Students are not permitted to use more than one answer book.*

*Students are not permitted to take the question paper away.*

*The following are provided:*
*A table of Fourier transforms*
*A Gaussian Tail Function graph*

Examiners responsible:

First Marker(s): A. Manikas
Second Marker(s): D. Mandic

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# PART-I

1. A matched filter is used to detect the signal $s(t)$, given by

$$\text{where } s(t) = 0.1 \text{ rect}\left\{\frac{t - 5\text{msec}}{10\text{msec}}\right\}$$

which is corrupted by additive white Gaussian noise with double-sided power spectral density $10^{-3}$ W/Hz. The maximum Signal-to-Noise (SNR) ratio at the filter output is [2 marks]

(a) 0.01;
(b) 0.1;
(c) 1;
(d) 10;
(e) none of the above.

2. The impulse response of an approximate-matched filter, matched (at time $T_o$) to the signal $\Lambda\left\{\frac{\tau}{T}\right\}$ in the presence of non-white noise of autocorrelation function $R_{nn}(\tau) = \Lambda\left\{\frac{\tau}{T}\right\}$ is [2 marks]

(a) $\delta(t)$;
(b) $\delta(t - T_o)$;
(c) $\text{sinc}^2(fT_o)$;
(d) $T_o\text{sinc}^2(fT_o)$;
(e) none of the above.

3. A matched filter is used to detect the signal $s(t)$, given by

$$s(t) = \begin{cases} t & \text{if } 0 \le t \le 20\text{ms} \\ 0 & \text{otherwise} \end{cases}$$

which is corrupted by additive white Gaussian noise with double-sided power spectral density $10^{-6}/3$ W/Hz. The maximum Signal-to-Noise (SNR) ratio at the filter output is [2 marks]

(a) 4
(b) 8
(c) 16
(d) 24
(e) None of the above.

4. Consider a binary pulse-code-modulation (binary-PCM) system where the digital modulation scheme being used is described as follows:

   "The input to the digital modulator is a binary sequence of 1's and 0's with the number of 1s being twice the number of zeros. The binary sequence is transmitted as a pulse signal $s(t)$ with a *one* being sent as $2.\text{rect}(\frac{t}{T_b}) + 4.\Lambda\left\{\frac{t}{T_b/2}\right\}$ and *zero* being sent as $0.\text{rect}\left\{\frac{t}{T_b}\right\}$."

   Furthermore, the channel noise is assumed to be additive and uniformly distributed between $-2$ Volts and $+2$ Volts. With $p_0(r)$ and $p_1(r)$ representing the likelihood functions of the above system which of the following is correct? [2 marks]

   (a) $p_0(r) = \frac{1}{12}\ \text{rect}\left\{\frac{r}{4}\right\}$, $\quad p_1(r) = \frac{1}{6}\ \text{rect}\left\{\frac{r-4}{4}\right\}$

   (b) $p_0(r) = \frac{1}{12}\ \text{rect}\left\{\frac{r}{4}\right\}$, $\quad p_1(r) = \frac{1}{6}\ \Lambda\left\{\frac{r-4}{4}\right\}$

   (c) $p_0(r) = \frac{1}{4}\ \text{rect}\left\{\frac{r}{4}\right\}$, $\quad p_1(r) = \frac{1}{4}\ \text{rect}\left\{\frac{r-4}{4}\right\}$

   (d) $p_0(r) = \frac{1}{4}\ \text{rect}\left\{\frac{r}{4}\right\}$, $\quad p_1(r) = \frac{1}{4}\ \Lambda\left\{\frac{r-4}{4}\right\}$

   (e) None of the above.

5. Consider a binary communication system which uses the following two equiprobable signals $s_0(t)$ and $s_1(t)$. The signals are transmitted over a communication channel which adds white Gaussian noise having a double-sided power spectral density of $10^{-6}$ W/Hz. The constellation diagram is given below, where the decision regions are also shown.

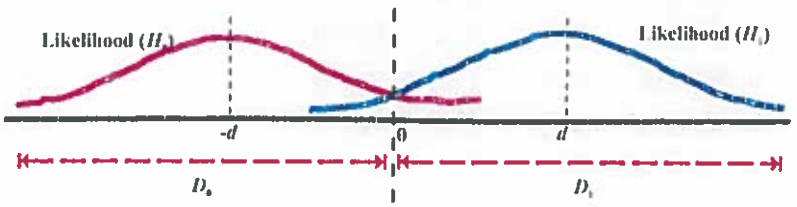



Figure Q10

If the forward transition matrix $\mathbb{F}$ of the equivalent discrete channel is

$$\mathbb{F} = \begin{bmatrix} 0.994. & 0.006 \\ 0.006. & 0.994 \end{bmatrix}$$

then [2 marks]

(a) $d = 12.5 \times 10^{-3}$;

(b) $d = 6.25 \times 10^{-3}$;

(c) $d = 2.5 \times 10^{-3}$;

(d) $d = 5 \times 10^{-3}$;

(e) none of the above.

6. The Fredholm integral equation of the first kind, which provides the general equation for a matched filter, is given as follows: [2 marks]

   (a) $\int_0^{T_{cs}} h_{\text{opt}}(\text{z}).R_{nn}(\tau-\text{z}).dz = s(T_{cs}-\tau)$ ; $0 \le \tau \le \infty$;
   (b) $\int_0^{T_{cs}} h_{\text{opt}}(\text{z}).R_{nn}(\tau-\text{z}).dz = s(T_{cs}-\tau)$ ; $0 \le \tau \le T_{cs}$;
   (c) $\int_0^{T_{cs}} h_{\text{opt}}(\text{z}).R_{nn}(\text{z}-\tau).dz = s(T_{cs}-\tau)$ ; $0 \le \tau \le \infty$;
   (d) $\int_0^{T_{cs}} h_{\text{opt}}(\text{z}).R_{nn}(\text{z}-\tau).dz = s(T_{cs}-\tau)$ ; $0 \le \tau \le T_{cs}$;
   (e) none of the above.

7. The minimum chip rate needed by a DS-BPSK spread spectrum system to resolve a multipath, with an additional path length of 30m compared to the direct path, is [2 marks]

   (a) 10 Mchips/sec;
   (b) 20 Mchips/sec;
   (c) 40 Mchips/sec;
   (d) 60 Mchips/sec;
   (e) none of the above.

8. Let $B_D = 16$MHz denotes the Doppler spread, $B_{\text{coh}}$ represents the coherent bandwidth and $T_c$ is the chip period. Then, in a frequency selective fast fading CDMA channel, which of the following is correct? [2 marks]

   (a) $T_c = 61n$ sec and $B_{\text{coh}} = 3$MHz.
   (b) $T_c = 61n$ sec and $B_{\text{coh}} = 100$MHz.
   (c) $T_c = 244n$ sec and $B_{\text{coh}} = 3$MHz.
   (d) $T_c = 244n$ sec and $B_{\text{coh}} = 100$MHz.
   (e) None of the above.

9. If the path-loss is 23.0103 dB and the power of the received signal is 6.9897 dBm, then the power of the transmitter is [2 marks]

   (a) 20 W;
   (b) 10 W;
   (c) 1 W;
   (d) 10 mW;
   (e) 20 mW.

10. With reference to 'multi-user (MU) CDMA receivers', which of the following statements is correct? [2 marks]

    (a) A RAKE receiver is a multi-user receiver.
    (b) A multi-user receiver is used to resolve paths (in a multipath environment), delayed by more than the chip period $T_c$.
    (c) A minimum-mse MU receiver requires no knowledge of the cross-correlation matrix of the PN-signals.
    (d) A decorrelating MU receiver is a sub-optimum multi-user receiver.
    (e) None of the above.

11. In a Space-Time DS-CDMA system employing a PN-code of length $N_c$ (with a chip period $T_c$) and an array of $N$ antennas at the receiver, which of the following statements involving the STAR manifold vector $\underline{h}$ of a path is correct? [2 marks]

    (a) $\|\underline{h}\|^2 = NN_c$
    (b) $\|\underline{h}\|^2 = N\ell$
    (c) $\|\underline{h}\|^2 = N(N_c - \ell)$
    (d) $\|\underline{h}\|^2 = N_c\ell$
    (e) None of the above.

    N.B.: The parameter $\ell \triangleq \left\lceil \frac{\tau}{T_c} \right\rceil \mathrm{mod} N_c$ is the discrete version of the path delay $\tau$.

12. Consider an antenna array system of 5 elements operating in the presence of one desired and two uncorrelated co-channel interfering signals, all of power $P_s = 1$. The power of the noise is $\sigma_n^2 = 10$. If $\mathbb{R}_{xx}$ is the theoretical covariance matrix of the received signal vector $\underline{x}(t)$, then which of the following statements is correct? [2 marks]

    (a) The minimum eigenvalue of $\mathbb{R}_{xx}$ is equal to 1.
    (b) The principal eigenvalue of $\mathbb{R}_{xx}$ is equal to 10.
    (c) The rank of $\mathbb{R}_{xx}$ is equal to 2.
    (d) The rank of $\mathbb{R}_{xx}$ is equal to 3.
    (e) The rank of $\mathbb{R}_{xx}$ is equal to 5.

13. Consider an antenna array system of $N$ elements operating in the presence $M$ co-channel sources ($M < N$). If $\underline{S}_i$ is the manifold vector associated with the $i^{th}$ source and $\mathbb{E}_s$ and $\mathbb{E}_n$ denote the matrices with columns the signal eigenvectors and the noise eigenvectors respectively of data covariance matrix $\mathbb{R}_{xx}$, then which of the following expresssions is correct? [2 marks]

    (a) $\mathbb{E}_n.\mathbb{E}_n^H.\underline{S}_i = \underline{S}_i$.
    (b) $\mathbb{E}_s.\mathbb{E}_s^H.\underline{S}_i = \underline{0}$.
    (c) $\mathbb{E}_n.\mathbb{E}_n^H.\underline{S}_i = \underline{0}$.
    (d) $\left(\mathbb{I}_N - \mathbb{E}_s.\mathbb{E}_s^H\right).\underline{S}_i = \underline{S}_i$.
    (e) None of the above.

14. Consider a MIMO wireless communication system where the Cartesian coordinates of the Tx and Rx antenna array elements are given by the columns of the following matrices

$$\text{Tx} : [\underline{\bar{r}}_1, \underline{\bar{r}}_2] = \begin{bmatrix} -1, & +2 \\ 0, & 0 \\ 0, & 0 \end{bmatrix} \text{ in units of half-wavelength,}$$

$$\text{Rx} : [\underline{r}_1, \underline{r}_2] = \begin{bmatrix} 0, & 0 \\ 0, & 0 \\ -0.5, & 0.5 \end{bmatrix} \text{ in units of half-wavelength.}$$

Which of the following statements, associated with its virtual Rx antenna array of an equivalent SIMO wireless communication system, is correct? [2 marks]

(a) $\begin{bmatrix} -1, & -1, & 2, & 2 \\ 0, & 0, & 0, & 0 \\ -0.5, & 0.5, & -0.5, & 0.5 \end{bmatrix}$.

(b) $\begin{bmatrix} -1, & 2, & -1, & 2 \\ -0.5, & -0.5, & 0.5, & 0.5 \\ 0, & 0, & 0, & 0 \end{bmatrix}$.

(c) $\begin{bmatrix} -0.5, & -0.5, & 0.5, & 0.5 \\ -1, & 2, & -1, & 2 \\ 0, & 0, & 0, & 0 \end{bmatrix}$.

(d) $\begin{bmatrix} -0.5, & -0.5, & 0.5, & 0.5 \\ 0, & 0, & 0, & 0 \\ -1, & 2, & -1, & 2 \end{bmatrix}$.

(e) None of the above.

15. Consider a linear array of 5 Rx-antennas having the following Cartesian coordinates:

$$[\underline{r}_1, \underline{r}_2, \underline{r}_3, \underline{r}_4, \underline{r}_5] = \begin{bmatrix} -5, & -1, & +1, & +2, & +3 \\ 0, & 0, & 0, & 0, & 0, \\ 0, & 0, & 0, & 0, & 0, \end{bmatrix} \text{ in units of half-wavelength.}$$

The rate of change of the arclength $\dot{s}(\theta)$ of the array manifold for a source with Direction-of-Arrival (azimuth) $\theta = 30^\circ$ is [2 marks]

(a) $\dot{s}(30^\circ) = 19.631$;

(b) $\dot{s}(30^\circ) = 9.9346$;

(c) $\dot{s}(30^\circ) = 5.4414$;

(d) $\dot{s}(30^\circ) = 3.1623$;

(e) none of the above.

16. Consider a beamformer which employs a uniform array of $N$ antennas and operates in the presence of a single signal with direction $(\theta = 30^0, \phi = 0^0)$. The carrier frequency is 2.4 GHz and the manifold vector for the Direction-of-Arrival $(\theta = 30^0, \phi = 0^0)$ is

$$[-0.1125 + 0.9936i, 0.6661 + 0.7458i, 1.0000, 0.6661 - 0.7458i, -0.1125 - 0.9936i]^T$$

Assume that the array steers its main lobe towards the direction $(\theta = 30^0, \phi = 0^0)$, the power of the received signal is 1 and the channel noise is additive white Gaussian noise of power 0.01. If at the output of the beamformer $P_{out}$ is the power of the desired signal and $\text{SNR}_{out}$ denotes the signal-to-noise ratio, which of the following statements is correct? [2 marks]

(a) $P_{out}$=5 and $\text{SNR}_{out}$=100.

(b) $P_{out}$=25 and $\text{SNR}_{out}$=100.

(c) $P_{out}$=5 and $\text{SNR}_{out}$=500.

(d) $P_{out}$=25 and $\text{SNR}_{out}$=500.

(e) None of the above.

17. For a uniform linear array of 5 sensors operating at 2.4GHz frequency with an inter-antenna spacing of 6.25cm, the beamwidth is [2 marks]

(a) 47.156°;

(b) 45.537°;

(c) 23.074°;

(d) 11.537°;

(e) none of the above.

18. For a Wiener-Hopf beamformer, which of the following statements is correct? [2 marks]

(a) It is a superresolution beamformer.

(b) It is robust to errors associated with the direction of the desired signal.

(c) It provides, asymptotically, complete interference cancellation.

(d) It is optimum with respect to SNIR criterion.

(e) None of the above.

19. Consider a beamformer which employes a uniform linear array of $N$ antennas and uses the following weight vector:

$$[-0.1125 + 0.9936i, 0.6661 + 0.7458i, 1.0000, 0.6661 - 0.7458i, -0.1125 - 0.9936i]^T.$$

If the channel noise is additive white Gaussian noise with power $\sigma_n^2 = 0.001$W then the noise power at the beamformer's output is [2 marks]

(a) 0.00025W;
(b) 0.0005W;
(c) 0.005W;
(d) 0.025W;
(e) none of the above.

20. Consider an array of 4 antennas with Cartesian coordinates given by the following matrix

$$\begin{bmatrix} -2, & 2, & 2, & -2 \\ -0.5, & -0.5, & 1, & 0.5 \\ 0, & 0, & 0, & 0 \end{bmatrix}$$ in units of half-wavelength.

The array aperture is [2 marks]

(a) 4.272;
(b) 4.0311;
(c) 4;
(d) 1.5;
(e) none of the above.

# PART-II

21. (a) The two signals $s_0(t)$ and $s_1(t)$ of a binary communication system each have energy equal to 93.3 and cross correlation coefficient 0.866.
    - Draw the constellation diagram of the system, properly labeled. [2 marks]
    - What is the distance between these two signals? [2 marks]

    (b) Define the concept of "Diversity". [3 marks]

    (c) Define the main four diversity combining rules. [3 marks]

22. Consider that one of the paths from the transmitter of a CDMA user arrives at the reference point of an antenna array CDMA receiver from direction (azimuth, elevation)= $(60°, 0°)$. The corresponding PN-sequence, of period $N_c$, is generated by the polynomial $D^2 + D + 1$ in GF(2) while the discrete path delay (mod-$N_c$) is equal to two. For this path, if the Cartesian coordinates of the antenna array elements are given by the columns of the following matrix

$$[r_1, r_2, r_3] = \begin{bmatrix} -1, & 0, & +1 \\ 0, & 0, & 0 \\ 0, & 0, & 0 \end{bmatrix} \text{ in units of half-wavelength,}$$

    find

    (a) the manifold vector; [5 marks]

    (b) the spatio-temporal array manifold vector. [5 marks]

23. Draw a block structure and write a mathematical equation for the impulse response of the following multipath frequency selective channels:

    (a) SISO, [2 marks]

    (b) SIMO, [2 marks]

    (c) MISO, and [2 marks]

    (d) MIMO. [2 marks]

24. Consider the signalling waveforms $s_1(t), s_2(t), s_3(t)$ and $s_4(t)$, as shown below




Figure Q24(a)

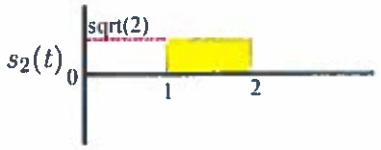



Figure Q24(b)

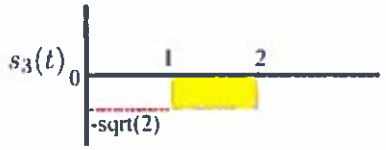



Figure Q24(c)

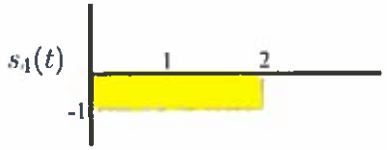



Figure Q24(d)

For this set of waveforms find:

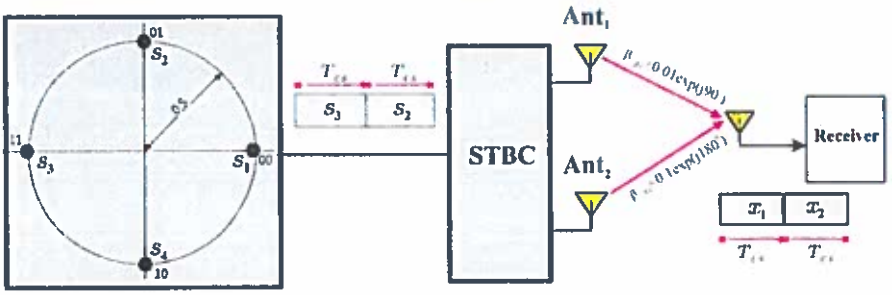
(a) the number $D$ of orthonormal signals $\{\varphi_i(t)\}$ that can be used to represent the above waveforms; [3 marks]

(b) the cross-correlation coefficient $\rho_{3,4}$: [2 marks]

(c) the constellation diagram; [3 marks]

(d) the minimum distance. [2 marks]

25. Consider the QPSK MISO system of 2 Tx antennas operating in a frequency-flat wireless channel as shown the following figure:




If the QPSK symbols $[s_3, s_2]$ are transmitted using the above "Space-Time Block Coder" (STBC), find the receiver's input $[x_1, x_2]$, ignoring the noise. [10 marks]

26. Consider an $M$-ary Communication System for which the signal set is described as follows:

$$s_i(t) = A_i \Lambda\left\{\frac{2t}{T_{cs}}\right\},\ i = 1, 2, \ldots, M.$$

$$\text{with} \begin{cases} M = 4 \\ A_i = (2i - 1 - M) \times 10^{-3}\text{Volts} \\ T_{cs} = 12\,\text{sec} \\ \Pr(\mathrm{H}_1) = \Pr(\mathrm{H}_4) = 1/8 \text{ and } \Pr(\mathrm{H}_2) = \Pr(\mathrm{H}_3) = 3/8 \end{cases}$$

The signals are transmitted over a communication channel which adds white Gaussian noise with a double-sided power spectral density of $10^{-6}$ W/Hz.

(a) Find the energy $E_i, i = 1, 2, 3, 4$. [3 marks]

(b) Calculate the values of the signal-vectors $\underline{w}_{s_i}$, $i = 1, 2, 3, 4$ for the above signal-set. [3 marks]

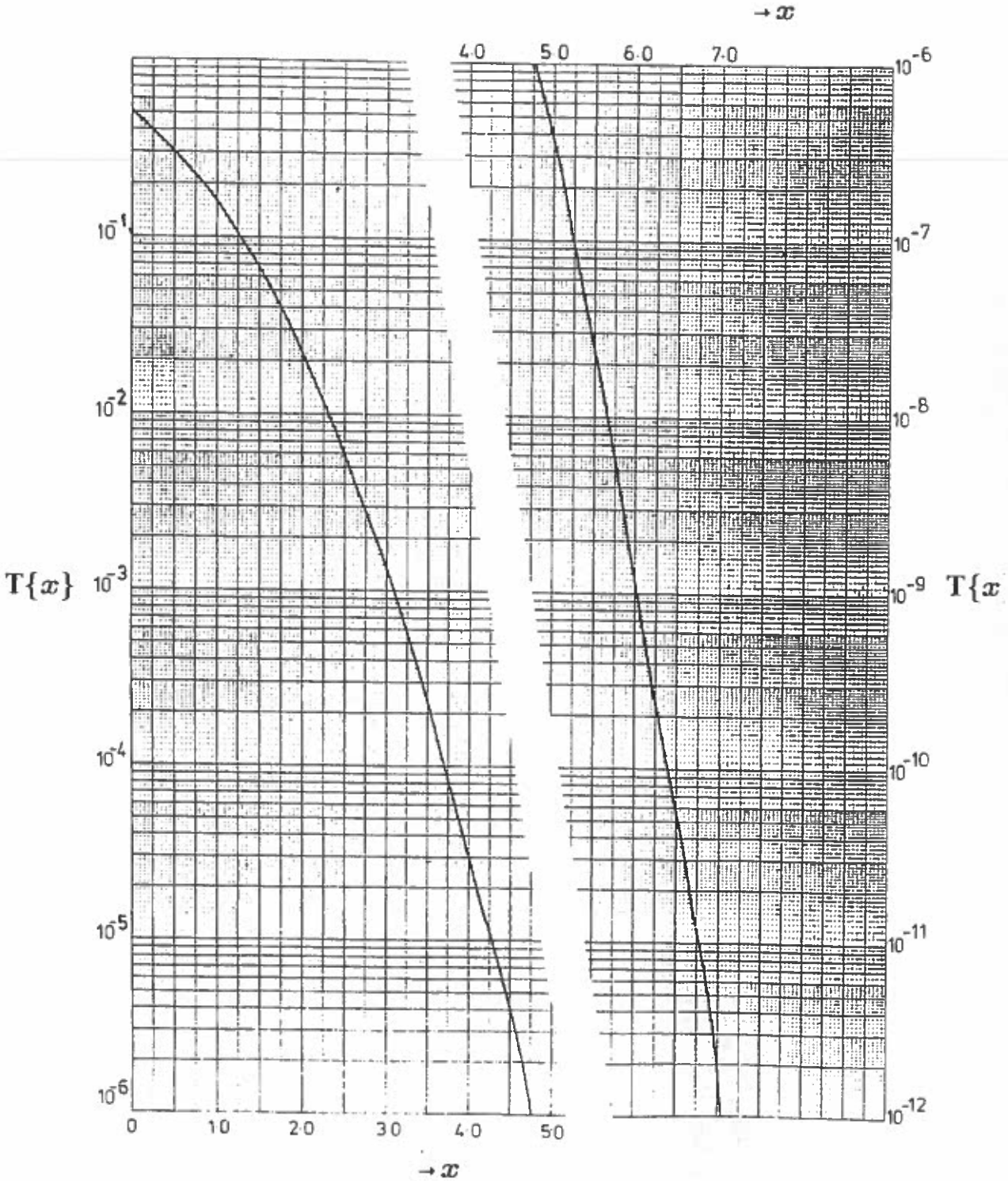
(c) Draw a labelled block diagram of the MAP receiver, based on the signals vectors $\underline{w}_{s_i}, i = 1,2,3,4$. [3 marks]

(d) Plot the constellation diagram and label the decision regions. [3 marks]

| Fourier Transform Tables | | | |
|---|---|---|---|
| | Description | Function | Transformation |
| 1 | Definition | $g(t)$ | $G(f) = \int_{-\infty}^{\infty} g(t).e^{-j2\pi ft}dt$ |
| 2 | Scaling | $g\left(\frac{t}{T}\right)$ | $\|T\|.G(fT)$ |
| 3 | Time shift | $g(t-T)$ | $G(f).e^{-j2\pi fT}$ |
| 4 | Frequency shift | $g(t).e^{j2\pi Ft}$ | $G(f-F)$ |
| 5 | Complex conjugate | $g^*(t)$ | $G^*(-f)$ |
| 6 | Temporal derivative | $\frac{d^n}{dt^n}g(t)$ | $(j2\pi f)^n.G(f)$ |
| 7 | Spectral derivative | $(-j2\pi t)^n.g(t)$ | $\frac{d^n}{df^n}G(f)$ |
| 8 | Reciprocity | $G(t)$ | $g(-f)$ |
| 9 | Linearity | $A.g(t) + B.h(t)$ | $A.G(f) + B.H(f)$ |
| 10 | Multiplication | $g(t).h(t)$ | $G(f) * H(f)$ |
| 11 | Convolution | $g(t) * h(t)$ | $G(f).H(f)$ |
| 12 | Delta function | $\delta(t)$ | $1$ |
| 13 | Constant | $1$ | $\delta(f)$ |
| 14 | Rectangular function | $\mathbf{rect}\{t\} \triangleq \begin{cases} 1 & \text{if } \|t\| < \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$ | $\mathbf{sinc}\{f\} \triangleq \frac{\sin(\pi f)}{\pi f}$ |
| 15 | Sinc function | $\mathbf{sinc}(t)$ | $\mathbf{rect}\{f\}$ |
| 16 | Unit step function | $u(t) \triangleq \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases}$ | $\frac{1}{2}\delta(f) - \frac{j}{2\pi f}$ |
| 17 | Signum function | $\mathrm{sgn}(t) \triangleq \begin{cases} 1 & t > 0 \\ -1 & t < 0 \end{cases}$ | $-\frac{j}{\pi f}$ |
| 18 | decaying exp (two-sided) | $e^{-\|t\|}$ | $\frac{2}{1+(2\pi f)^2}$ |
| 19 | decaying exp (one-sided) | $e^{-\|t\|}.u(t)$ | $\frac{1-j2\pi f}{1+(2\pi f)^2}$ |
| 20 | Gaussian function | $e^{-\pi t^2}$ | $e^{-\pi f^2}$ |
| 21 | Lambda function | $\Lambda\{t\} \triangleq \begin{cases} 1-t & \text{if } 0 \le t \le 1 \\ 1+t & \text{if } -1 \le t \le 0 \end{cases}$ | $\mathbf{sinc}^2\{f\}$ |
| 22 | Repeated function | $\mathbf{rep}_T\{g(t)\} = g(t) * \mathbf{rep}_T\{\delta(t)\}$ | $\left\|\frac{1}{T}\right\|\mathbf{comb}_{\frac{1}{T}}\{G(f)\}$ |
| 23 | Sampled function | $\mathbf{comb}_T\{g(t)\} = g(t).\mathbf{rep}_T\{\delta(t)\}$ | $\left\|\frac{1}{T}\right\|\mathbf{rep}_{\frac{1}{T}}\{G(f)\}$ |

The graph below shows the Tail function $\mathbf{T}\{x\}$ which represents the area from $x$ to $\infty$ of the Gaussian probability density function N(0,1), i.e.

$$\mathbf{T}\{x\} = \int_x^{\infty} \frac{1}{\sqrt{2\pi}} . \exp\left(-\frac{y^2}{2}\right) dy$$




Note that if $x > 6.5$ then $\mathbf{T}\{x\}$ may be approximated by $\mathbf{T}\{x\} \approx \frac{1}{\sqrt{2\pi}.x} . \exp\left\{-\frac{x^2}{2}\right\}$